Machine Learning Fundamentals

Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn from data and improve their performance without being explicitly programmed. Unlike traditional programming, where instructions are written explicitly, ML algorithms learn rules directly from data.

Key Concepts

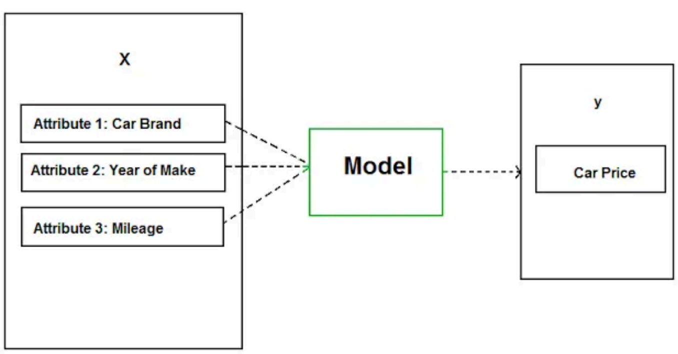
- **Training Data**: Examples the model learns from.
- **Algorithms**: Mathematical methods that learn from data.
- **Models:** Patterns learned by algorithms used to make predictions.
- **Predictions:** Outcomes generated by the trained model.

Types of Machine Learning

Machine learning methods are commonly grouped into **three main types** based on how they learn from data: **Supervised**, **Unsupervised**, and **Reinforcement Learning**. Each type addresses a different class of problems and learning scenarios.

1. Supervised Learning

Supervised learning involves algorithms that learn from labeled datasets, meaning each data point has an associated output. The goal is to learn a general rule mapping inputs to outputs, so that the model can predict the correct output for new, unseen inputs. Essentially, the model is "supervised" by the known answers during training.



Supervised Learning Model

Source: Medium

A- Classification

Classification – Predicting a **category** or class label for each input.